

Johnny Rhe

SungKyunkwan University: 2066, Seobu-ro, Jangan-gu, Suwon-si, Gyeonggi-do, Korea

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Research Interests

I am a Ph.D. candidate in the Department of Electrical and Computer Engineering at Sungkyunkwan University. My research focuses on **hardware-software (HW-SW) co-design for efficient AI systems**, with an emphasis on hardware-aware optimization and energy-efficient acceleration for on-device AI. My work explores **optimization across the model, mapping, and dataflow levels**, particularly for Compute-In-Memory (CIM) architectures. More recently, I have expanded my research to modern AI workloads, including Vision Transformers (ViTs) and Large Language Models (LLMs), with the goal of enabling scalable and energy-efficient AI computing systems.

Education

B.S. School of Electronic and Electrical Engineering

Mar. 2011 - Feb. 2019

SungKyunkwan University (SKKU), Suwon, Korea

Ph.D. Electrical and Computer Engineering

Sept. 2019 – Present

Sungkyunkwan University (SKKU), Suwon, Korea

- Intelligent & Resource-Efficient Image Processing & Systems Design (IRIS) Lab
- **Thesis Advisor: Prof. Jong Hwan Ko**

Visiting Researcher

Mar. 2026 – Present

Purdue University, Indiana, USA

- Nanoelectronics Research Laboratory (NRL) under the supervision of Prof. Kaushik Roy
-  [H18]  [G3] Financial Support from SKKU

Research Experience

Accelerated Computation Algorithms for CIM-based D/CNN Inference

Sept. 2019 - Present

- Developed novel DNN weight mapping methods [C2, C15]
- Proposed multi-level mapping optimization via pruning [J2, C4, J5, C5, J3]
- Proposed low-rank and kernel grouping methods for compact weight representation [C11, C7]
- Proposed input-stationary mapping methods for efficient and flexible inference [C3, J4]
- Developed an algorithm to increase partial-sum sparsity in ADC-less CIM architectures
- Developed an input-skipping algorithm for efficient CIM-based convolution operation

Reliability Enhancement on Analog CIM Arrays

Sept. 2022 - 2024

- Designed fault-robust remapping schemes to mitigate inference error [C5]
- Proposed hybrid grouping to improve multi-bit weight precision and fault tolerance [C12]

Software-Based Simulator for CIM-based Inference

Sept. 2021 - 2023

- Designed visualization and evaluation frameworks integrated with NeuroSim [C9]

Resource-Efficient Neural Networks for Edge Applications

Sept. 2023 - 2024

- Proposed lightweight CNN/SNN models for biomedical signal decoding [C13, C8]

Projects

A Study on Collaborative Mapping and Training Techniques for Energy-Efficient In-Memory CNN Inference

Sept. 2024 - Sept. 2025

National Research Foundation (NRF), Korea

-  [G1] Project Leader

Optimizing DNN Mapping and Execution on In-Memory Computing Arrays

April. 2024 - Feb. 2026

Ministry of Trade, Industry & Energy (MOTIE), Korea

HW-SW Co-Design for Low-Power Real-Time In-Memory Deep Learning Operation

Mar. 2023 - Feb. 2026

Samsung Advanced Institute of Technology (SAIT), Korea

Spatiotemporal Precision Control for Efficient and Robust In-Memory Deep Learning Inference

National Research Foundation (NRF), Korea

June. 2023 - May. 2024

A Co-Optimization Simulation Framework for Efficient In-Memory Deep Learning Computation

National Research Foundation (NRF), Korea

June. 2022 - May. 2023

Energy-Efficient DNN Design on Memristor-Based IC platforms

Institute for Information & communications Technology Planning & Evaluation (IITP), Korea

Sept. 2020 - Mar. 2023

Honors & Awards

[H19] Selected for the DAC Young Fellows Program

Design Automation Conference (DAC) 2026

April, 2026

[H18] Global Value Innovation Creator (GVIC) Program Selection

Sungkyunkwan University (SKKU)

Dec, 2025

[H17] Oral and Poster Presentation at Student Research Competition

International Conference on Computer-Aided Design (ICCAD) 2025

Dec, 2025

[H16] 3rd Place Winners at Grand Challenge on Brain-Machine Interfaces

IEEE Biomedical Circuits and Systems Conference (BioCAS)

Oct, 2025

[H15] 2025 SKKU Innovative Research Fellowship

Sungkyunkwan University (SKKU)

May, 2025

[H14] Poster Presentation at PhD Forum

Design, Automation and Test in Europe Conference (DATE) 2025

Mar, 2025

[H13] Outstanding Research Award

Korea Collaborative & High-tech Initiative for Prospective Semiconductor Research (K-CHIPS)

Nov, 2024

[H12] KETEP Outstanding Research Award

Korea Institute of Energy Technology Evaluation and Planning (KETEP)

Oct, 2024

[H11] Samsung Outstanding Paper Award - Samsung Elec. & SKKU (2024)

Samsung Electronics

Aug, 2024

[H10] Grand Prize at the Exynos AI Challenger

Samsung Electronics' S.LSI

July, 2024

[H9] 3rd Place Winner in the System Design Contest (GPU Track)

Design Automation Conference (DAC) 2024

June, 2024

[H8] Poster Presentation at PhD Forum

IEEE International Symposium on Circuits and Systems (ISCAS) 2024

May, 2024

[H7] Most Popular Poster Award at the Student Research Forum

29th Asia South Pacific Design Automation Conference (ASP-DAC) 2024

Jan, 2024

[H6] Poster Presentation at Student Research Forum

29th Asia South Pacific Design Automation Conference (ASP-DAC) 2024

Jan, 2024

[H5] Samsung Outstanding Paper Award - Samsung Elec. & SKKU (2023)

Samsung Electronics

Aug, 2023

[H4] IEEE CASS Non-Flagship Conference Student Travel Grant (Selection) - ISLPED

IEEE Circuits and Systems Society (CAS)

Aug, 2023

[H3] Best Paper Award

Korean Artificial Intelligence Association (KAIA)

July, 2023

[H2] 2022 SKKU Innovative Research Fellowship

Sungkyunkwan University (SKKU)

Sept, 2022

[H1] 2nd Place Winner of Artificial Intelligence Grand Challenge

Institute of Information & Communications Technology Planning & Evaluation, Korea

July, 2020

Featured Publications

JOURNAL ARTICLES

[J5] **Johnny Rhe**, Kang Eun Jeon and Jong Hwan Ko, “Genetic Algorithm-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping,” *Journal of Systems Architecture (JSA)*, May. 2025 (JCR Q1). [[Paper](#)]

[J3] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seong Moon Jeong, and Jong Hwan Ko, “KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping,” *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)* vol. 71, no. 12, pp. 6138–6151, Nov. 2024 (JCR Q1). [[Paper](#)]

🏆 **[H12] Outstanding Research Award from KETEP (2024)**

[J2] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, “VWC-SDK: Convolutional Weight Mapping Using Shifted and Duplicated Kernel with Variable Windows and Channels,” *IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)* vol. 12, no. 2, pp. 408–421, May. 2022 (JCR Q1). [[Paper](#)]

CONFERENCE PROCEEDINGS

[C15] **Johnny Rhe**, Juhong Park, Kang Eun Jeon, and Jong Hwan Ko, “ETA: Efficient Transformer Attention Mapping for ReRAM-based Compute-In-Memory Architectures,” *Asia Pacific Conference on Circuits and Systems (APCCAS)*, Oct. 2025. [[Paper](#)]

[C5] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seong Moon Jeong, and Jong Hwan Ko, “Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays,” *ACM/IEEE International Conference on Computer-Aided Design (ICCAD)*, Oct. 2023. [[Paper](#)]

🏆 **[H11] Outstanding Paper Award from Samsung Elec.(2024)**

[C4] **Johnny Rhe**, Kang Eun Jeon, and Jong Hwan Ko, “PAIRS: Pruning-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping in Processing-In-Memory Architectures,” *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, Aug. 2023. [[Paper](#)]

🏆 **[H4] Student Travel Grant Selection from IEEE CAS**

[C2] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, “VW-SDK: Efficient Convolutional Weight Mapping Using Variable Windows for Processing-In-Memory Architectures,” *Design, Automation, and Test in Europe (DATE)*, Mar. 2022 (Oral, acceptance rate: 25%). [[Paper](#)]

Full Publications (160+ Citations)

Most-up-to-date publication list can be founded at [Google Scholar](#)

JOURNAL ARTICLES

[J5] **Johnny Rhe**, Kang Eun Jeon and Jong Hwan Ko, “Genetic Algorithm-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping,” *Journal of Systems Architecture (JSA)*, May. 2025 (JCR Q1). [[Paper](#)]

[J4] Juhong Park, **Johnny Rhe**, and Jong Hwan Ko, “Input/Mapping Precision Controllable Digital PIM with Adaptive Adder Tree Architecture for Flexible DNN Inference,” *Journal of Systems Architecture (JSA)*, Feb. 2025 (JCR Q1). [[Paper](#)]

[J3] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seong Moon Jeong, and Jong Hwan Ko, “KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping,” *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)* vol. 71, no. 12, pp. 6138–6151, Nov. 2024 (JCR Q1). [[Paper](#)]

🏆 [H12] Outstanding Research Award from KETEP (2024)

[J2] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, “VWC-SDK: Convolutional Weight Mapping Using Shifted and Duplicated Kernel with Variable Windows and Channels,” *IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)* vol. 12, no. 2, pp. 408–421, May. 2022 (JCR Q1). [[Paper](#)]

[J1] Eunyong Lee, Taeyoung Han, Donguk Seo, Gicheol Shin, Jaerok Kim, Seonho Kim, Soyoun Jeong, **Johnny Rhe**, Jaeyung Park, Jong Hwan Ko, Yoonmyung Lee, “A Charge-Domain Scalable-Weight In-Memory Computing Macro with Dual-SRAM Architecture for Precision-Scalable DNN Accelerators,” *IEEE Transactions on Circuits and Systems I (TCAS-I)* vol. 68, no. 8, pp. 3305–3316, May. 2021 (JCR Q1). [[Paper](#)]

CONFERENCE PROCEEDINGS

[C15] **Johnny Rhe**, Juhong Park, Kang Eun Jeon, and Jong Hwan Ko, “ETA: Efficient Transformer Attention Mapping for ReRAM-based Compute-In-Memory Architectures,” *Asia Pacific Conference on Circuits and Systems (APCCAS)*, Oct. 2025. [[Paper](#)]

[C14] **Johnny Rhe**, and Jong Hwan Ko, “Weight Sharing for Array-Efficient CNN Inference in Compute-In-Memory Architectures,” *International SoC Design Conference (ISOCC)*, Oct. 2025. [[Paper](#)]

[C13] Chanwook Hwang, Chan Woong Park, **Johnny Rhe**, and Jong Hwan Ko, “Efficient and Robust SNN Decoder Training for Closed-Loop Brain-Machine Interfaces,” *IEEE Biomedical Circuits and Systems Conference (BioCAS)*, Oct. 2025. [[Paper](#)]

🏆 [H16] 3rd Place Winner at Grand Challenge by BioCAS

[C12] Kang Eun Jeon, Sangheum Yeon, Jinhee Kim, Hyeonsu Bang, **Johnny Rhe**, and Jong Hwan Ko, “Row-Column Hybrid Grouping for Fault-Resilient Multi-Bit Weight Representation on IMC Arrays,” *ACM/IEEE International Conference on Computer-Aided Design (ICCAD)*, Oct. 2025. [[Paper](#)]

[C11] Kang Eun Jeon, **Johnny Rhe**, and Jong Hwan Ko, “Low-Rank Compression for IMC Arrays,” *Design, Automation & Test in Europe Conference (DATE)*, Mar. 2025. [[Paper](#)]

[C10] **Johnny Rhe**, and Jong Hwan Ko, “Row-Efficient Pruning for In-Memory Convolutional Weight Mapping,” *International SoC Design Conference (ISOCC)*, Aug. 2024. [[Paper](#)]

[C9] Kang Eun Jeon¹, Wooram Seo¹, **Johnny Rhe**, and Jong Hwan Ko, “ConvMapSim: Modeling and Simulating Convolutional Network Mapping on PIM Arrays,” *IEEE Artificial Intelligence Circuits and Systems (AICAS)*, Apr. 2024. [[Paper](#)]

[C8] Chanwook Hwang, Jaehyeon So, **Johnny Rhe**, Jiyeon Kim, Juhong Park, Kang Eun Jeon, and Jong Hwan Ko, “An Efficient Ventricular Arrhythmias Detection on Microcontrollers with Optimized 1D CNN,” *IEEE Artificial Intelligence Circuits and Systems (AICAS)*, Apr. 2024. [[Paper](#)]

🏆 [H10] Grand Prize at Exynos AI Challenger by Samsung Elec.

[C7] Juhong Park, **Johnny Rhe**, and Jong Hwan Ko, “KARS: Kernel-Grouping Aided Row-Skipping for SDK-based Weight Compression in PIM Arrays,” *IEEE International Symposium on Circuits and Systems (ISCAS)*, May 2024. [[Paper](#)]

🏆 [H13] Best Paper Award from KAIA (2023)

[C6] Hyeonsu Bang, Kang Eun Jeon, **Johnny Rhe**, and Jong Hwan Ko, “DCR: Decomposition-Aware Column Re-Mapping for Stuck-At-Fault Tolerance in ReRAM Arrays,” *IEEE International Conference on Computer Design (ICCD)*, Nov. 2023. [[Paper](#)]

[C5] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seong Moon Jeong, and Jong Hwan Ko, “Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays,” *ACM/IEEE International Conference on Computer-Aided Design (ICCAD)*, Oct. 2023. [[Paper](#)]

🏆 [H11] Outstanding Paper Award from Samsung Elec. (2024)

[C4] **Johnny Rhe**, Kang Eun Jeon, and Jong Hwan Ko, “PAIRS: Pruning-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping in Processing-In-Memory Architectures,” *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, Aug. 2023. [[Paper](#)]

🏆 [H4] Student Travel Grant Selection from IEEE CAS

[C3] Kang Eun Jeon, **Johnny Rhe**, and Jong Hwan Ko, “Weight-Aware Activation Mapping for Energy-Efficient Convolution on PIM Arrays,” *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, Aug. 2023. [[Paper](#)]

🏆 [H5] Outstanding Paper Award from Samsung Elec. (2023)

[C2] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, “VW-SDK: Efficient Convolutional Weight Mapping Using Variable Windows for Processing-In-Memory Architectures,” *Design, Automation, and Test in Europe (DATE)*, Mar. 2022 (Oral, acceptance rate: 25%). [\[Paper\]](#)

[C1] Gicheol Shin, Donguk Seo, Jaerok Kim, **Johnny Rhe**, Eunyong Lee, Seonho Kim, Soyoun Jeong, Jong Hwan Ko, Yoonmyung Lee, “A charge-Domain Computation-in-Memory Macro with Versatile All-Around-Wire-Capacitor for Variable-Precision Computation and Array-Embedded DA/AD Conversions,” *European Conference on Solid-State Circuits (ESSCIRC)*, Sept. 2021. [\[Paper\]](#)

Patents

[P10] **Johnny Rhe**, and Jong Hwan Ko, KR Patent No.10-2999338, Apparatus and method for controlling process-in-memory by accelerating convolution operation based on arranging pattern of weight in kernel, and storage medium storing instructions to perform method for controlling process-in-memory, July. 2026

[P9] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, US Patent No.12,670,366, Memory device for optimizing computation of convolution layer, method for controlling memory device, and recording medium storing instruction to perform method for controlling memory device, June. 2026

[P8] **Johnny Rhe**, Kang Eun Jeon, and Jong Hwan Ko, US Patent Application (with Samsung Electronics) No.19/069,410, Method and electronic device with weight pruning, Mar. 2025

[P7] Juhong Park, **Johnny Rhe**, and Jong Hwan Ko, US Patent Application (with Samsung Electronics) No.18/932,587, Device and method with processing-in-memory, Oct. 2024

[P6] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, and Jong Hwan Ko, PCT Patent Application No.PC-T/KR2024/016234, Method and apparatus for convolution operation utilizing kernel shape control, Oct. 2024

[P5] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, KR Patent No.10-2710011, Memory device for optimizing computation of convolution layer, method for controlling memory device, and recording medium storing instruction to perform method for controlling memory device, Sept. 2024

[P4] **Johnny Rhe**, Kang Eun Jeon, and Jong Hwan Ko, KR Patent Application (with Samsung Electronics) No.10-2024-0058360, Pattern-based weight pruning method and electronic devices, May. 2024

[P3] Juhong Park, **Johnny Rhe**, and Jong Hwan Ko, KR Patent Application (with Samsung Electronics) No.10-2024-0008936, Device and method with processing-in-memory, Jan. 2024

[P2] **Johnny Rhe**, and Jong Hwan Ko, US Patent Application No.18/398,389, Apparatus and method for controlling process-in-memory by accelerating convolution operation based on arranging pattern of weight in kernel, and storage medium storing instructions to perform method for controlling process-in-memory, Dec. 2023

[P1] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, and Jong Hwan Ko, KR Patent Application No.10-2023-0171519, Method and apparatus for convolution operation utilizing kernel shape control, Nov. 2023

Talks

Optimizing Weight Mapping for Energy-Efficient In-Memory CNN Inference Oct, 2025
International Conference on Computer-Aided Design (ICCAD)

ETA: Efficient Transformer Attention Mapping for ReRAM-based Compute-In-Memory Architectures Oct, 2025
Asia Pacific Conference on Circuits and Systems (APCCAS)

Enabling Resource-Efficient AI via HW-SW Co-Design Aug, 2025
University of Illinois Chicago (UIC)

Enabling Resource-Efficient AI via HW-SW Co-Design Aug, 2025
Duke University

Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays Oct, 2023
International Conference on Computer Aided Design (ICCAD)

Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays Sept, 2023
Sungkyunkwan University (SKKU) AI Colloquium

PAIRS: Pruning-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping in Processing-In-Memory Architectures Aug, 2023
International Symposium on Low Power Electronics and Design (ISLPED)

VWC-SDK: Convolutional Weight Mapping Using Shifted and Duplicated Kernel with Variable Windows and Channels Sept, 2022
Sungkyunkwan University (SKKU) AI Colloquium

VW-SDK: Efficient Convolutional Weight Mapping Using Variable Windows for Processing-In-Memory Architectures Mar, 2022
Design, Automation, and Test in Europe (DATE)

Teaching Experience

Teaching Assistant in Operating System
Sungkyunkwan University (SKKU), Suwon, Korea

- Instructor: Prof. Jong Hwan Ko

Mar, 2020

Student Guidance

Mentor for Undergraduate Research Program
Sungkyunkwan University (SKKU), Suwon, Korea

- Research on Layer-wise Window Skipping for CNN Acceleration
- Jihoon Jeong, hongjun ju, jungjae Lee

Dec. 2025 - Jan. 2026

Mentor for Undergraduate Research Program
Sungkyunkwan University (SKKU), Suwon, Korea

- Dynamic Computation Adjustment Techniques for IMC-based Efficient CNN Inference
- Yehyeon Kim, Seongjo Lee

Sept. 2025 - Present

Graduate Research Mentorship
Sungkyunkwan University (SKKU), Suwon, Korea

- Kernel-Grouping Techniques for IMC Arrays [C7] - 🏆 [H12] Best Paper Award from KAIA (2023)
- Juhong Park

Mar. 2023 - Mar. 2024

Mentor for Outstanding Undergraduate Research Program
Sungkyunkwan University (SKKU), Suwon, Korea

- Remapping Method for Mitigating Cell Variation in IMC Arrays
- Minho Lee, Minseop Kim, Seonghoon Jeong, Jihoon Choi

Feb. 2022 - June. 2022

Grants

Development of Application Processor Design Technologies for Multimodal Sensor
Signal Processing
Sungkyunkwan University (SKKU), Suwon, Korea

- Contributed to concept development and writing of the proposal
- The Ministry of Trade, Industry and Energy(MOTIE), Korea (Project # RS-2026-25521189)

April, 2026

Global Value Innovation Creator (GVIC) Program
Sungkyunkwan University (SKKU), Suwon, Korea

- Contributed to concept development and writing of the proposal
- Conducted visiting research at Purdue University
- Supported by SKKU
- Amount awarded: ₩ 30.0 million (\$ 21 thousand)

Jan, 2026

Development of Collaborative Intelligent Agent Technologies for Human-AI
Symbiotic Cooperation
Sungkyunkwan University (SKKU), Suwon, Korea

- Contributed to concept development and writing of the proposal
- Institute for Information & communications Technology Planning & Evaluation (IITP), Korea (Project # RS-2025-25442569)
- Amount awarded: ₩ 11.0 billion (\$ 8.15 million)

July, 2025

A Study on Collaborative Mapping and Training Techniques for Energy-Efficient
In-Memory CNN Inference
Sungkyunkwan University (SKKU), Suwon, Korea

- Core idea development and proposal writing as a project leader
- National Research Foundation (NRF), Korea (Project # RS-2024-00409161)
- Amount awarded: ₩ 25 million (\$ 17.9 thousand)

July, 2024

Technical Reviewer Activities

IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)
Design Automation Conference (DAC)

Skills

Programming Python, PyTorch, VHDL, Verilog
Tools NeuroSim

References

Dr. Jong Hwan Ko (*Ph.D. advisor*)

Associate Professor, *Dept. of Electrical and Computer Engineering*,
Sungkyunkwan University (SKKU), Suwon, Korea
🏠 <https://iris.skku.edu/> ✉ jhko@skku.edu

Dr. Kang Eun Jeon

Postdoctoral Researcher, *Kim Jaechul Graduate School of Artificial Intelligence*,
Korea Advanced Institute of Science & Technology (KAIST), Seongnam, Korea
🏠 [Google Scholar](#) ✉ kejeon@kaist.ac.kr